What is a megawatt-hour of solar worth?

The estimand

The capture rate of solar in the DE-LU bidding zone, by calendar year 2019 to 2025: the generation-weighted average day-ahead price solar earns, divided by the baseload average price. Then the slope of that rate against solar's share of load, with onshore wind as the comparator.

The answer

Solar's capture rate fell from 92.8% in 2019 to 51.4% in 2025 while its share of load rose from 8.4% to 15.1%. Onshore wind went from 85.8% to 86.5% at a share of 19.9% to 22.6%.

Slope for solar: -5.4 capture-rate points per point of share. 95% interval [-9.9, -0.8] across the 7 years (least squares, 5 degrees of freedom); [-6.6, -4.1] if only the weeks of each year are resampled.

Wind: +0.3 [-2.0, +2.5]; the interval holds zero.

What it does not say

The slope is a description of 7 years. Used to forecast, it loses to last year's value: backtest RMSE 0.25 against 0.15 (capture-rate units). The 20% share scenario carries a backtest interval of [-0.42, 0.98], which is no information.

Solar's capture rate is falling. How fast it keeps falling is not something this data can fix.

Method

Data

Energy-Charts API (Fraunhofer ISE; data from Bundesnetzagentur SMARD.de, CC BY 4.0), fetched by src/fetch.py month by month: /price (DE-LU day-ahead) and /public_power (Germany, quarter-hourly Solar, Wind onshore, Load). Complete years only; no quarter-hour is missing in any year. 2026 is partial (to 31 Oct) and is used once, as a further out-of-sample test.

Capture rate

CR = [sum(g x p) / sum(g)] / mean(p), over every quarter-hour of the calendar year (Berlin time). Prices are hourly until 30 Sep 2025 and quarter-hourly after; an hourly price is held across its four quarters, which equals the roadmap's hourly average of generation joined to the hourly price. Applying it literally changes 2025 by +0.13 points.

Intervals

Block bootstrap over 7-day blocks, 2,000 draws, seed 15. Weeks are resampled inside each year and calendar quarter, so the season mix is fixed. One draw of weeks serves solar, wind, baseload and load at once: the error they share is drawn once. Other years are never resampled.
This measures how much each annual figure depends on which weeks occurred. It does not see the scatter of the 7 years around the line, so I also report the least-squares interval (5 df) and a leave-one-year-out check.

Backtest

Slope fitted on years up to t-1 predicts the capture rate of year t at its realised share of load, t = 2022 to 2025, against two benchmarks (last year's rate; the mean so far). The share itself is not forecast. A fifth test uses January to October 2026 against the same window of earlier years.

Sweep

Negative prices kept or floored; price resolution; the 2022 spike kept or dropped; start year; functional form; bootstrap block length and stratification.

Result

Annual capture rates

2019: 93% 2020: 81% 2021: 78% 2022: 94% (solar)

2023: 76% 2024: 59% 2025: 51% (solar)

Wind runs between 71% and 87%. Baseload 2019: 38 EUR/MWh; 2022: 235; 2025: 89.

Hours at a negative price: 211 in 2019, 575 in 2025; 24% of 2025 solar output fell in them.

The slope

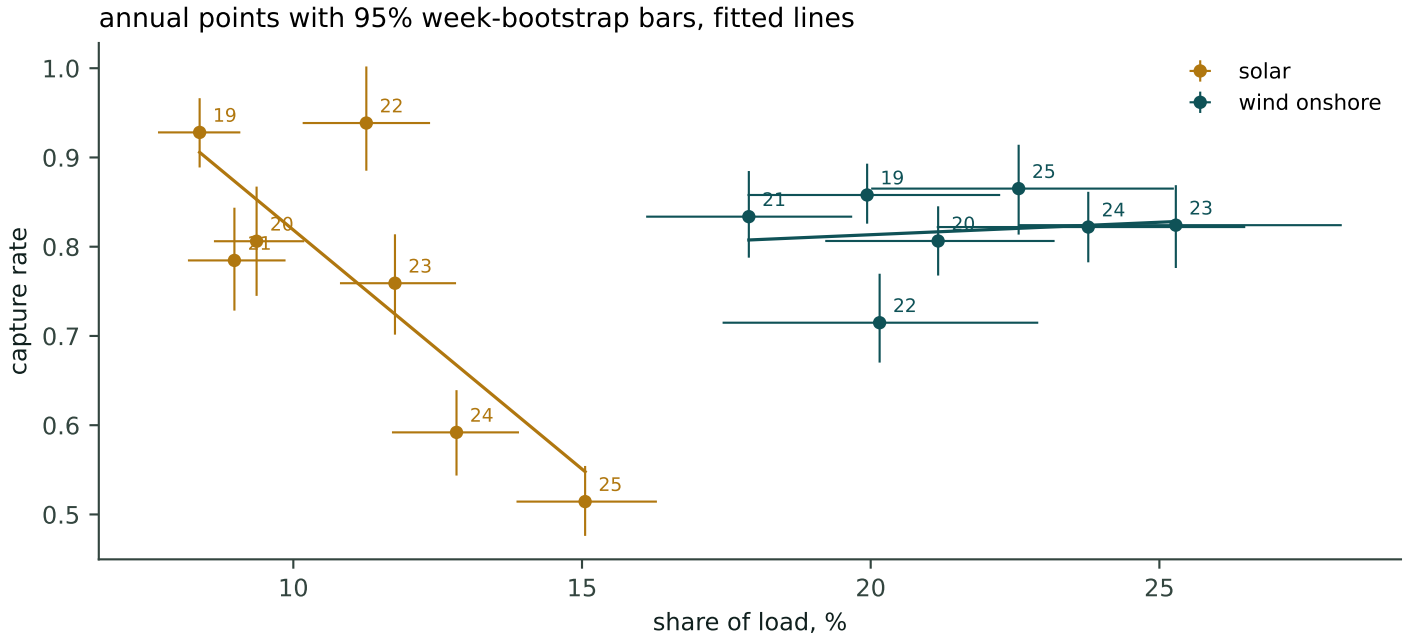
Solar -5.36 points per point of share (bootstrap standard error 0.64, least-squares 1.78).

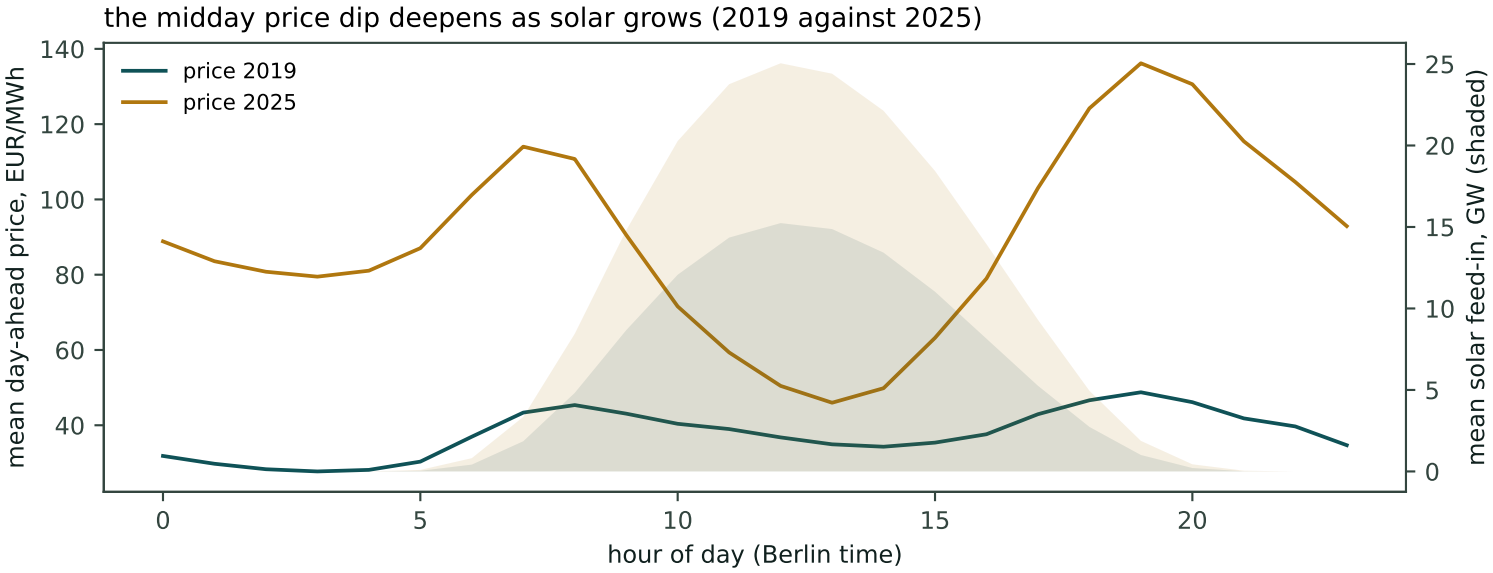
Leave-one-year-out slopes run from -6.12 to -4.37: no single year drives the sign.

Solar minus wind: -5.64; bootstrap [-6.97, -3.96], leave-one-year-out [-9.76, -1.52].

Both exclude zero: solar's rate falls with its own share and wind's does not.

Within a year, weekly rate on weekly share (year effects) gives -1.83: a different, short-run relation.





Out of sample: the slope does not forecast

Backtest, solar, 2022 to 2025

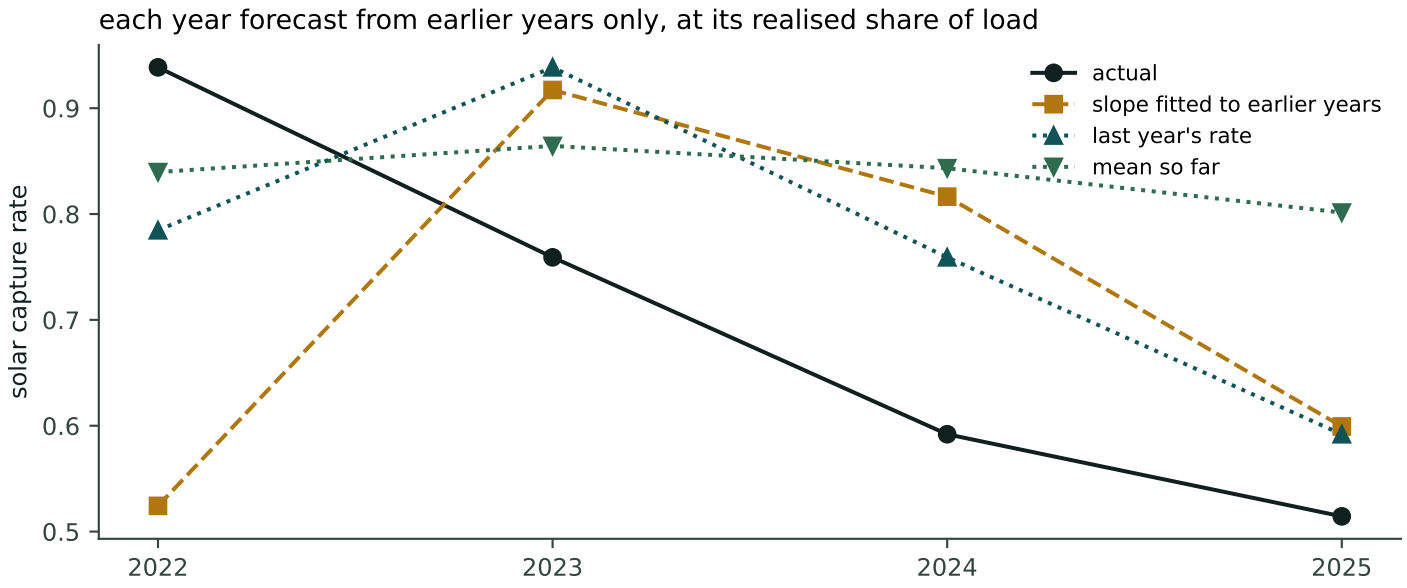
RMSE in capture-rate units: slope model 0.252, last year 0.150, mean so far 0.204.
Without 2022: 0.166, 0.149, 0.228. The slope model is not better than the naive rule in either.
Wind: slope model 0.080, last year 0.084, mean so far 0.066; nothing to forecast beyond the mean.

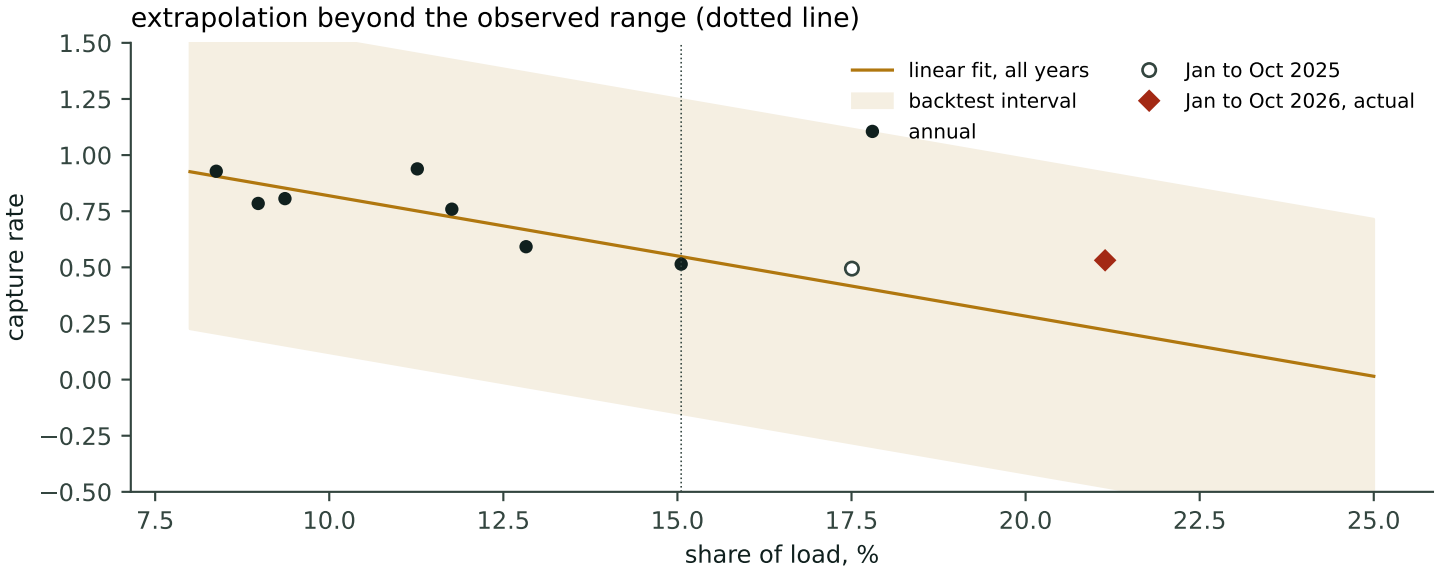
Fifth test, January to October 2026

Share of load 21.1% (Jan to Oct 2025: 17.5%). Fitted line predicts 0.32; realised 0.53;
last year's window 0.49. The rate stopped falling while the share rose, so the line overshoots.

The scenario, and why its interval is empty

At 20.1% of load (15.1% in 2025 plus 5 points, a value I chose) the line gives 0.28; log-linear 0.37.
Backtest interval [-0.42, 0.98] (RMSE x 2.776, four errors). The parameter-only bootstrap, [0.16, 0.40],
ignores model error and is the wrong one to quote. Observed shares stop at 15.1%.





Validation and sensitivity

Against the published annual market values (2025)

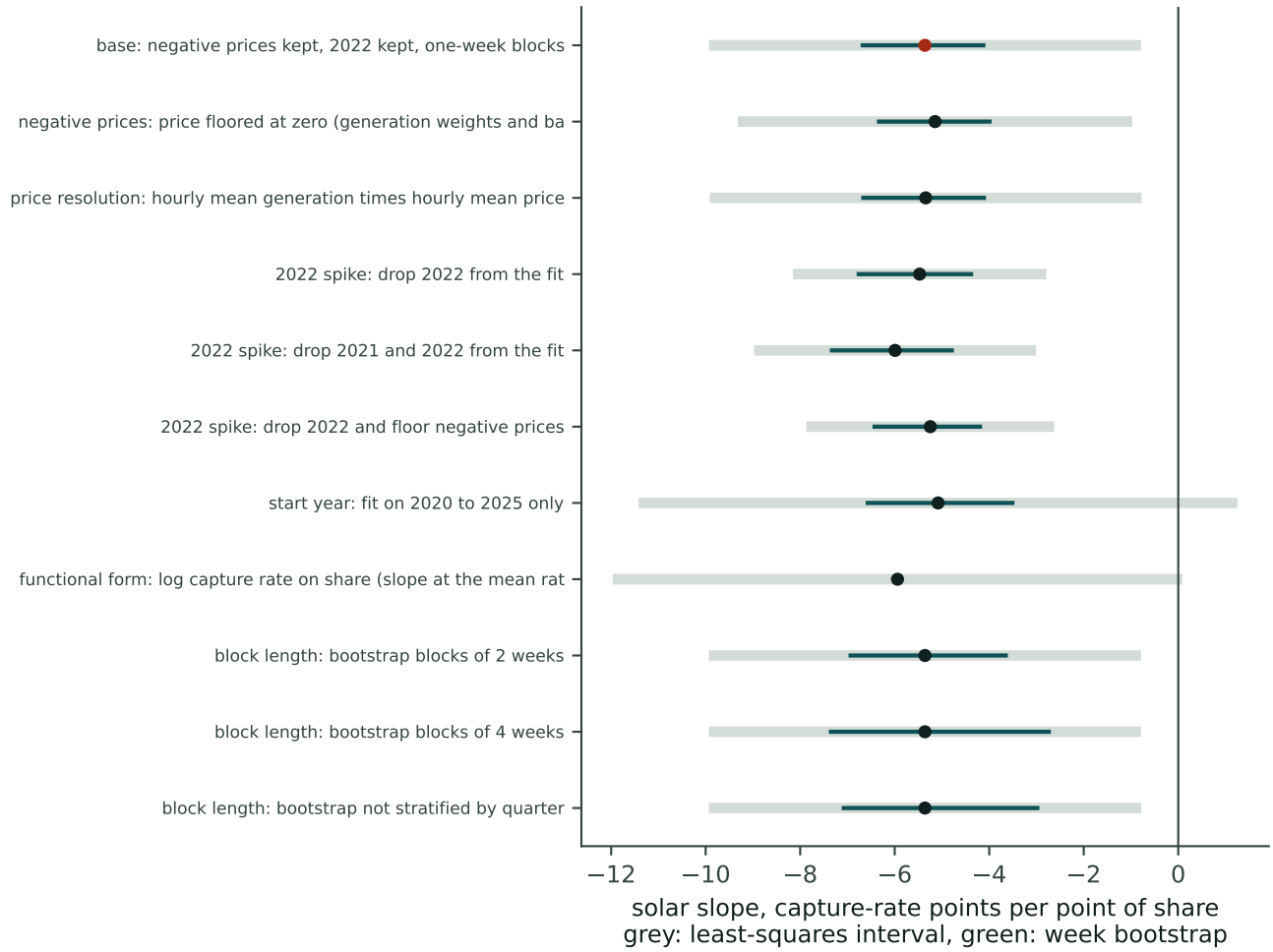
The transmission system operators publish annual market values (JW) under the EEG. For 2025: baseload 89.32 EUR/MWh published, 89.32 mine; solar 45.08 against 45.95 (+1.9%); onshore wind 74.41 against 77.27 (+3.8%). The page gives only the current year, so this is one year, not a series. It checks the price base and the weighting, not the slope.

A published series that does not match

The TSOs' MW Solar(a) (EEG 2012 s.33) exists for every year but differs from my solar-weighted price by 15% on average and by up to 26% (2025), mine lower in 6 of 7 years. Levels correlate at 0.99. I could not establish its weighting and do not use it as a check; I report the gap.

What moves the solar slope

Across 11 variants it runs from -5.99 to -5.08; negative in all, but the least-squares interval reaches zero or above in 2 of them (start year; functional form). Dropping 2022: -5.47. Flooring negative prices: -5.15.
Wind runs from -0.37 to +0.52; its intervals hold zero in every variant that has them.
The bootstrap block length moves the interval, not the point: width 4.6 with 4-week blocks against 2.6 with one week.



Limits, and what I would build next

The assumption that breaks it

That the share of load is what moves the capture rate. The seven annual points span a low baseload price in 2020 (30 EUR/MWh), the spike of 2022 (94% capture at 11.3% share) and the fall after it, and the slope is not separated from them. Storage build-out, rules on feed-in at negative prices and the move to quarter-hourly prices are changes I did not measure that fall in the same window.
The backtest shows what that costs: errors of both signs, up to 0.41 in capture-rate units.

What a sceptical reviewer attacks first

Seven points. The week bootstrap gives [-6.6, -4.1] and the across-year interval [-9.9, -0.8]; the second is the one to quote. A capture rate is a ratio of prices, so a change in the gas price moves numerator and denominator together only if the whole curve shifts in proportion; it need not. Generation is the published feed-in, net of self-consumption. Load is grid load.

Other limits

One bidding zone, day-ahead only: intraday and balancing revenues, and what a project realises under a PPA or the market premium, are outside the estimand. The validation covers one year of the TSO values; the MW Solar(a) series disagrees with me and I could not resolve it from public pages.

Next

Residual-load capture rate (price on load net of wind and solar) to separate supply shape from level; monthly panel with gas and carbon prices as controls; the same rate for the Netherlands and Spain.